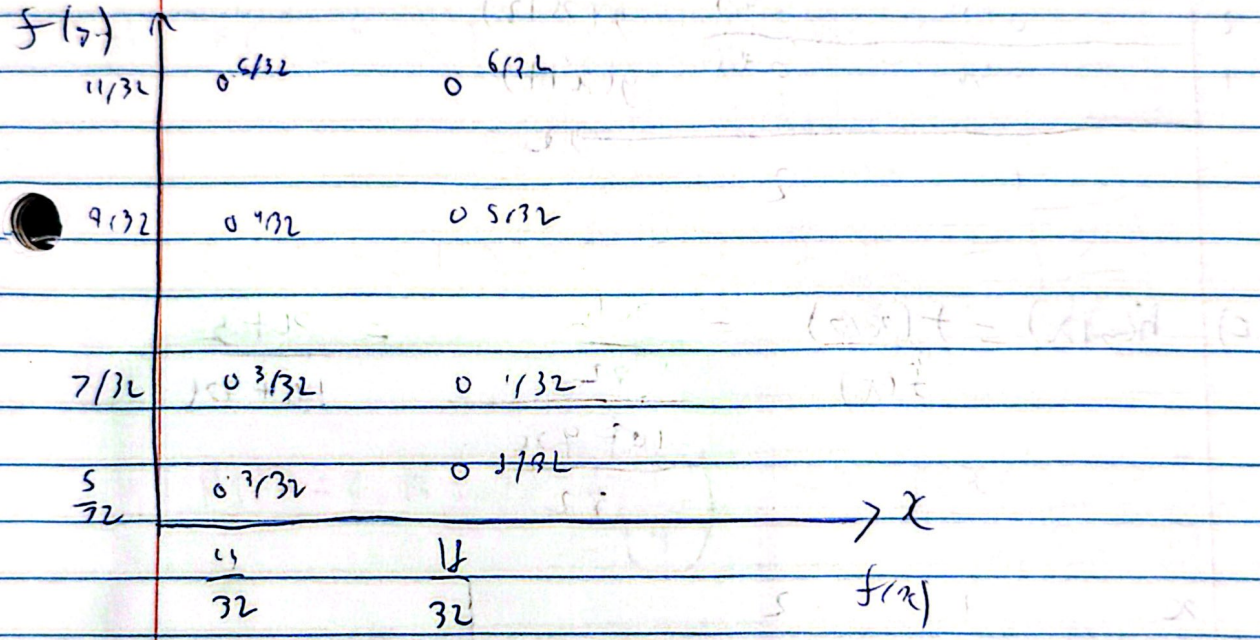
HW 10

4.3-1

$f(x,y) = \frac{x+y}{32}$ $\quad x=1,2 \quad y=1,2,3,4$

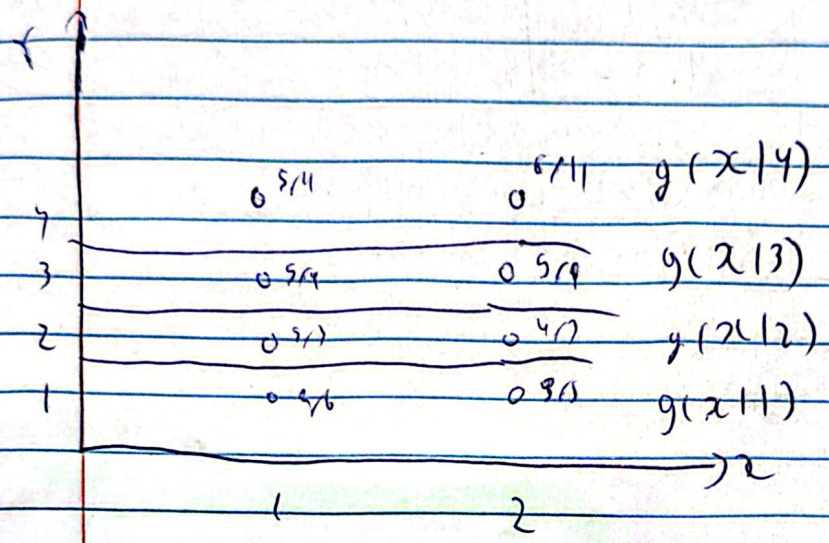
x / y	1	2	P(Y=y)
1	2/32	3/32	5/32
2	3/32	4/32	7/32
3	4/32	5/32	9/32
4	5/32	6/32	11/32
P(X=x)	14/32	18/32	1



b) $g(x|y) = \frac{f(x,y)}{f(y)} = \frac{\frac{x+y}{32}}{\frac{3+2y}{32}}$

$g(x|y) = \frac{x+y}{3+2y}$ $\quad x=1,2 \quad y=1,2,3,4$

x	1	2
y		
1	2/5	3/5
2	3/7	4/7
3	4/9	5/9
4	5/11	6/11



c) $h(y|x) = \frac{f(x|y)}{f(x)} = \frac{\frac{x+y}{72}}{\frac{10+4x}{72}} = \frac{x+y}{10+4x}$

x	1	2
y		
1	2/14	3/18
2	3/14	4/18
3	4/14	5/18
4	5/14	6/18

1 2

y	h(y|1)	h(y|2)
4	0 5/14	0 0/18
3	0 4/14	0 5/18
2	0 3/14	0 4/18
1	0 2/14	0 3/18
	1	2

d) $P(1 \le Y \le 3 \mid X=1) = \sum_{y=1}^{3} \frac{1+2}{14} = \boxed{\frac{9}{14}}$

$P(Y \le 2 \mid X=2) = \frac{3}{18} + \frac{4}{18} = \boxed{\frac{7}{18}}$

$P(X=2 \mid Y=3) = \boxed{\frac{5}{9}} \rightarrow \frac{P(X=2, Y=3)}{P(Y=3)}$

e) $E(Y \mid X=1) = \sum_{i} y_i \frac{P(Y=y_i, X=1)}{P(X=1)}$

$$= \frac{\frac{2}{32} + \frac{6}{32} + \frac{12}{32} + \frac{20}{32}}{\frac{14}{32}} = \frac{40}{14}$$

$\boxed{E(Y \mid X=1) = 2.8571}$

$$E(Y^2 \mid X=1) = \sum_i y_i^2 \frac{P(Y=y_i, X=1)}{P(X=1)}$$

$$= \frac{\frac{2}{32} + 4\left(\frac{3}{32}\right) + 9\left(\frac{4}{32}\right) + 16\left(\frac{5}{32}\right)}{\frac{14}{32}}$$

$$= 130/14$$

$$E(Y^2 \mid X=1) = 9.2857$$

$$Var(Y \mid X=1) = E(Y^2 \mid X=1) - (E(Y \mid X=1))^2$$

$$\frac{130}{14} - \frac{1600}{196} = 1.1225$$

4.3 - 2

$(1,1) = \frac{3}{8}$, $(2,1) = \frac{1}{8}$, $(1,2) = \frac{1}{8}$, $(2,2) = \frac{3}{8}$

$P(X=1) = P(X=2) = P(Y=1) = P(Y=2) = \frac{1}{2}$

$P(X=1 \mid Y=1) = \frac{3}{4}$, $P(X=2 \mid Y=1) = \frac{1}{4}$

$P(X=1 \mid Y=2) = \frac{1}{4}$, $P(X=2 \mid Y=2) = \frac{3}{4}$

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$Y=1 \quad E(X|1) = \frac{5}{4} \quad E(X^2|1) = \frac{7}{4} \quad Var(X|1) = \frac{3}{16}$

$Y=2 \quad E(X|2) = \frac{7}{4} \quad E(X^2|2) = \frac{13}{4} \quad Var(X|2) = \frac{3}{16}$

Conditional

$P(Y=1 \mid X=1) = \frac{3}{4}, \quad P(Y=2 \mid X=1) = \frac{1}{4}$

$P(Y=1 \mid X=2) = \frac{1}{4}, \quad P(Y=2 \mid X=2) = \frac{3}{4}$

$E[Y \mid X=1] = \frac{5}{4}, \quad E[Y \mid X=2] = \frac{7}{4}, \quad Var(Y \mid X) = \frac{3}{16}$

5.1 - 6

a) $F(x) = \int_0^x f(t)\,dt = \int_0^x 2t\,dt = 2\left[\frac{t^2}{2}\right]_0^x$

$= x^2$

$$F(x) = \begin{cases} 0 & x \le 0 \\ x^2 & 0 < x < 1 \\ 1 & x \ge 1 \end{cases}$$

b) $R = F(x) = x^2 \qquad F(x) = R \qquad x^2 = R$

$x = \sqrt{R} \quad \rightarrow \quad U \sim Uniform(0,1), \quad x = \sqrt{R}$

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(0,1)

$R=0.7 \quad x=\sqrt{0.7} \approx 0.8367$

$R=0.4 \approx 0.6325$

$x=\sqrt{0.9} \approx 0.9487$

$R=0.6 \approx 0.7746$

$R=0.1 \approx 0.3162$

$R=0.2 \approx 0.4472$

$R=0.16 = 0.4$

$R=0.8 \approx 0.8944$

$R=0.09 = 0.3$

$R=0.21 \approx 0.4583$

$x: \{0.8367, 0.6325, 0.9487, 0.7746, 0.3162, 0.4472, 0.4, 0.8944, 0.3, 0.4583\}$

5.1-7

$x \to pdf \to f_x(x) = \theta x^{\theta-1}, \quad 0 < x < 1$

Let $y = -2\theta \ln x$, then: $x = e^{-y/(2\theta)}$,

$\frac{dx}{dy} = \frac{-1}{2\theta} e^{-y/(2\theta)}$

$f_Y(y) = \frac{1}{2} e^{-y/2}, \quad y > 0$

$Y \sim Exp\left(\lambda = \frac{1}{2}\right)$

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5.1-12

$X \sim U(-1,3) \rightarrow f_X(x) = 1/4$ on $(-1,3)$. Let $Y = X^2$, so $0 \le y \le 9$. For $0 \le y \le 1$, both roots are in support, so

$$f_Y(y) = \frac{1}{4}\left(\frac{1}{2\sqrt{y}} + \frac{1}{2\sqrt{y}}\right) = \frac{1}{4\sqrt{y}}$$

For $1 < y \le 9$, only $+\sqrt{y}$ is in support

$$f_Y(y) = \frac{1}{8\sqrt{y}}, \text{ else } f_Y(y) = 0$$

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